Playtest Feedback Platform

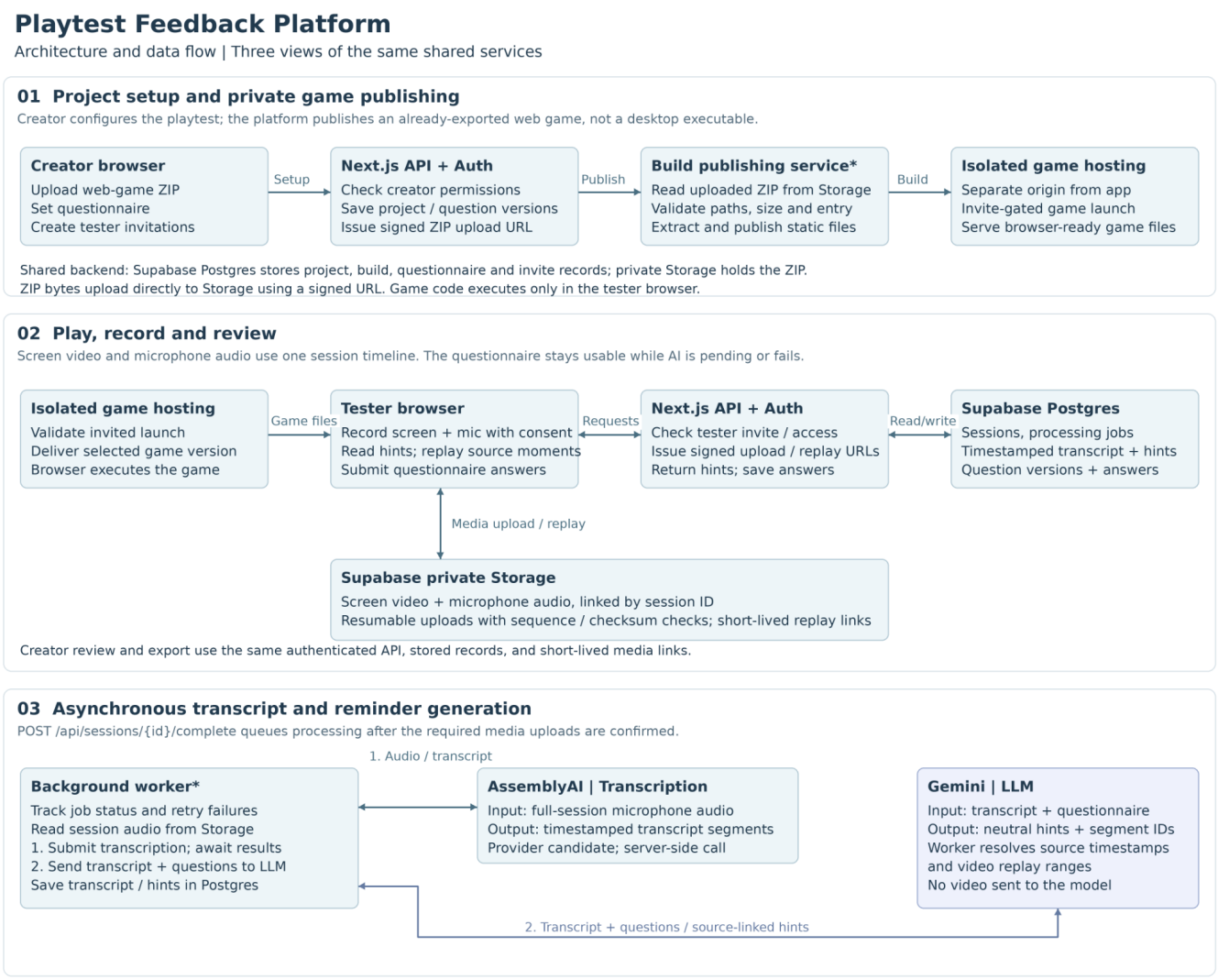
TECHNICAL SPECIFICATION | CS146S AUT26 | PAPRIKA CHEN & AMILIA LIU | OCTOBER 2026

Private browser playtests with timestamped feedback

Creators upload a web-game ZIP, create a questionnaire, and invite testers. The platform privately publishes browser-ready files without offering the original ZIP as a tester download. During play, it records screen video and microphone audio on one timeline. After the session, a timestamped transcript feeds Gemini prompts under relevant questionnaire questions, each with a button to replay the linked video moment. Session media, transcript, questionnaire, and answers are available to download for 7 days, then automatically deleted.

Stack: TypeScript, Next.js, React, Supabase JavaScript client, browser MediaRecorder APIs, AssemblyAI pre-recorded transcription API, and Gemini API.

Architecture and data flow



* Proposed services: build publishing and background processing runtimes / deployment host remain to be selected.

Current spec: session media and derived data expire after 7 days; builds remain until creator deletion.

Repeated service names refer to the same infrastructure. All provider credentials remain server-side.

Figure 1. Creator setup, private game publishing, session capture, review, and asynchronous transcript/reminder processing. Build-publishing and background-processing runtimes remain proposed implementation services.

Success metrics and implementation reference

Pilot results should include participant counts. Use findings to set numerical targets for later iterations rather than inventing thresholds before the first pilot.

1. Pilot success metrics

Metric	How to measure
Creator workflow completion	Share of creators who upload a supported game, configure a questionnaire, and generate a working invite without help. Track failure by step.
Tester workflow completion	Share of testers who start a session, complete recording, and submit the questionnaire. Track drop-off and technical failures by step.
Reminder usefulness	Ask testers to rate helpfulness from 1-5. Review a sample for question relevance and faithfulness to the transcript.
Processing wait time	Measure session completion to reminders available; report median, maximum, and failures.
Replay accuracy	For a manually checked sample, verify replay starts within 1 second of the source utterance; report the share that meets the target.

2. Core API

JSON over HTTPS via Next.js Route Handlers. Auth checks creator permissions and tester invite tokens. Upload media in resumable chunks; use idempotency keys for session creation and recording completion.

Route	Key inputs and outputs
POST /api/projects	Project name/settings → project ID.
POST /api/projects/{id}/builds	ZIP metadata → build ID and private upload URL; validate and publish build.
POST /api/projects/{id}/questionnaires	Questionnaire title/questions → questionnaire ID and version.
POST /api/projects/{id}/invites	Build, questionnaire, invitees → scoped invite links/status.
POST /api/sessions	Invite token/consent → session ID and recording setup.
POST /api/sessions/{id}/recording-chunks	Screen/audio chunk, sequence, checksum → upload status.
POST /api/sessions/{id}/complete	Final chunk IDs → processing status; queue transcript then Gemini prompts.
GET /api/sessions/{id}/recall-prompts	Question IDs, hints, source timestamps, replay ranges.
POST /api/sessions/{id}/answers	Question ID/edited answer → saved answer ID.
GET /api/projects/{id}/feedback	Sessions, answers, transcripts, and processing status.
GET /api/sessions/{id}/export	Within 7 days: download video, audio, transcript, questionnaire, and answers.
DELETE /api/sessions/{id}	Delete all session data immediately; scheduled deletion at day 7.

3. Security and reliability

Get consent before screen/audio capture. Validate ZIP paths, file count, and uncompressed size; never execute game code on the server. Serve builds from an isolated origin and gate launches by invite. Use short-lived authenticated download links. Confirm AssemblyAI/Gemini data-retention and training terms: platform deletion after 7 days does not control third-party copies. Test denied permissions, timestamp drift, interrupted uploads, invalid ZIPs, processing failures, replay accuracy, expiry/deletion, and cross-project access. Keep questionnaire submission usable when AI is pending or fails.

4. Agent-ready project

Keep database migrations, generated TypeScript types, a sample game/questionnaire, API contracts, and acceptance criteria in the repo. CI runs formatting, lint/type checks, unit/API tests, and browser end-to-end tests for creator setup and tester play, recording, survey, and review.